

UNDERSTANDING WHILE LOOPS IN PYTHON

A basic guide to using while loops for iteration in
Python



WHAT IS A WHILE LOOP?

- A 'while' loop repeatedly executes a block of code as long as the condition is True.
- The syntax is:
 - while condition:
 - # Code to execute
- The loop stops when the condition becomes False.



BASIC EXAMPLE OF A WHILE LOOP

- # Example:
- counter = 1
- while counter <= 5:
- print(counter)
- counter += 1 # Increment the counter
- Output: 1 2 3 4 5



INFINITE LOOP

- A while loop can become infinite if the condition never becomes False.
- Example:
 - while True:
 - print("This will run forever unless stopped!")
- To avoid infinite loops, ensure that the condition will eventually become False.



BREAKING A WHILE LOOP

- You can break a while loop using the 'break' statement.
- # Example:
 - counter = 1
 - while True:
 - if counter > 5:
 - break
 - print(counter)
 - counter += 1
- # Output: 1 2 3 4 5



COMMON USE CASES OF WHILE LOOPS

- 1. Counting or iterating through a series of numbers.
- 2. Continuously prompting for user input until a valid input is received.
- 3. Implementing game loops or repeating tasks until a condition is met.



SUMMARY

- - A 'while' loop runs as long as the condition is True.
- - It is important to make sure the loop will eventually stop.
- - Use 'break' to exit a while loop early.
- - Common use cases include counting, input validation, and game loops.

